For international companies: available via legal entities in Cyprus, Georgia, or Dubai. Deel also works.

Arkadiy Pechnikov

Tech Lead | Python & Rust | Crypto Infrastructure & Trading Systems

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PROFESSIONAL SUMMARY

Engineering leader who still writes code. As founding CTO, built a 9-person team from zero and delivered a crypto trading platform on $100K - roughly a tenth of what competitors spent. Architected the SDK generation pipeline (mitmproxy → OpenAPI → auto-SDK) that turned 20+ undocumented exchange APIs into a week's work each instead of a month. Migrated a broken monolith to Temporal.io, pushing throughput 20x to 200 jobs/sec. Comfortable hiring, managing budgets, running code reviews, and jumping into Ghidra to reverse-engineer a WASM binary when the team needs it. Looking for a lead role where I can build teams and still shape the architecture.

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript/TypeScript, SQL
Leadership & Process: Team Building (up to 9), Code Review, Architecture Decisions, Hiring, Budget Management, Agile
Frameworks & Libraries: FastAPI, aiohttp, asyncio, Celery, Temporal.io, Pydantic, SQLAlchemy
Cloud & DevOps: Kubernetes, Docker, FluxCD, GitHub Actions, CI/CD, Helm, Cloudflare, MinIO, NixOS, k3s, Terraform
Databases: PostgreSQL, Redis, MySQL, Elasticsearch, TimescaleDB
Observability: Grafana, Victoria Metrics, Loki, Sentry
Blockchain & DeFi: Web3.py, ethers.js, EVM, DeFi protocols (Uniswap, 1Inch, Stargate, LayerZero), TON SDK, mitmproxy
Security & Analysis: Web Application Analysis (Burp Suite, Chrome DevTools), JavaScript Deobfuscation, WebSocket Protocol Analysis, On-chain Transaction Analysis, ABI Encoding/Decoding, Traffic Analysis (mitmproxy), WASM Binary Analysis (Ghidra)

PROFESSIONAL EXPERIENCE

Lead Backend Engineer

Fenixwb - Wildberries slot booking service - 200 jobs/sec, 10K+ users | Oct 2024 - Sep 2025 | Remote

- Inherited an unstable monolith prototype pushing ~10 tasks/sec - redesigned the entire backend on Temporal.io with ~10 microservices integrated with Elasticsearch and MySQL
- Pushed throughput from 10 to 200 background jobs/sec (20x), the threshold that let the product handle real user load and start onboarding paying customers
- Disassembled a WASM captcha binary in Ghidra, then rewrote the solver in Rust - execution dropped from 2s to 0.1s (20x faster), giving clients higher booking success rates than competing bots
- Migrated from MongoDB to PostgreSQL - simpler queries, fewer ops headaches for a 3-person backend team
- Sourced and hired 2 engineers (backend + DevOps) from scratch, building the engineering function from zero - owned architecture calls, task allocation, and code review
- Went from zero monitoring to production-grade observability: Grafana + Victoria Metrics + Loki with custom dashboards and alerting - incidents detected in ~3 minutes vs previously hours, MTTR dropped from days to hours
- Chose to scrap the failing monolith for Temporal.io over incremental patches - the architectural call that enabled 10→200 tasks/sec across ~10 services

Tech: Rust, Python, Temporal.io, Elasticsearch, MySQL, PostgreSQL, asyncio, Grafana, Victoria Metrics, Loki, Docker, Kubernetes

Founding CTO / Technical Lead

Cryptosofters - Multi-wallet orchestration platform with visual pipeline builder - near-production, drag-and-drop wallet operations | May 2023 - Nov 2024 | Remote (Dubai-based)

- Designed multi-wallet orchestration platform with drag-and-drop pipeline constructor - users configured chains of on-chain operations (swaps, bridges, contract calls) without code. Built to near-production before investment stopped
- Under the hood: event-driven state machine with per-wallet operation queues, automatic retry with exponential backoff, and transaction rollback on partial failures - the engine that made drag-and-drop orchestration reliable enough for real money
- Zero security incidents across the platform's lifetime. Architected per-user wallet isolation, credential management, and transaction signing in a multi-tenant environment where investor confidence hinged on it
- Reverse-engineered closed-source smart contracts via on-chain transaction analysis and ABI decoding - often the only way to integrate protocols with no SDK, no docs, and no public API
- Architected the SDK generation pipeline: mitmproxy traffic capture → OpenAPI spec → auto-generated Python SDK for 20+ CEX APIs. Cut exchange integration from ~1 month to ~1 week (75%) - the force multiplier that made a 9-person team viable on a $100K budget

- Delivered 20+ DeFi protocol integrations (Uniswap, 1Inch, Stargate, LayerZero) with a unified interface via web3.py - per-protocol time dropped from ~1 week to ~1 day (~80%), and any team member could add protocols, not just the specialists
- 9 engineers (4 backend, 3 frontend, 1 Solidity, 1 PM), $100K annual budget, zero turnover - kept the team together through a budget crisis that would've scattered most startups
- Grew the team from 0 to 9: wrote job descriptions, ran technical interviews, onboarded everyone, and shaped engineering culture from day one
- Set up code review standards, CI/CD pipelines, sprint cadence, and knowledge-sharing sessions - bug escape rate dropped and delivery became predictable enough to commit to external timelines
- Runway hit zero. Assembled an investor pitch deck, led the fundraising conversations myself, and bought the team more time - the moment I learned operational discipline matters as much as architecture

Tech: Python, Web3.py, mitmproxy, OpenAPI, asyncio, PostgreSQL, Redis, Docker

Python Developer / Tech Lead

Vitrinagram - Telegram MiniApp e-commerce platform - multi-tenant storefronts, 1,000+ concurrent users per store | Nov 2021 - Apr 2023 | Remote (Moscow-based)

- Built multi-tenant Telegram MiniApp backends handling 1,000+ concurrent users per store - horizontal scaling via Celery + Kubernetes meant cost grew linearly with tenants, not exponentially
- Automated tenant provisioning: FluxCD + Cloudflare API pipeline, ~3 minutes from commit to live subdomain
- Per-tenant observability out of the box: Grafana + Victoria Metrics + Loki + Sentry dashboards clients could use for self-service debugging - ~40% fewer support tickets once clients could self-debug
- Led backend development across a team of 7, owning architecture decisions for multi-tenant scalability, setting technical direction for the platform, and mentoring mid-level engineers through code review and design discussions that improved team velocity over two quarters
- Spotted a technical leadership vacuum that was slowing delivery - escalated to the CEO, which led to team restructuring and a noticeable improvement in sprint completion rates

Tech: Python, FastAPI, Celery, PostgreSQL, Redis, Docker, Kubernetes, FluxCD, Grafana, Victoria Metrics, Loki, Sentry, MinIO, Cloudflare

Python Backend Developer / Lead

Freelance - Crypto infrastructure, trading systems, and data pipelines | Jan 2018 - Present

- Built EVM wallet management system for a client with multiple investors: 2,000+ wallets, 5,000+ tx/day, fully autonomous - equivalent of eliminating a full-time ops person
- Led 3 Go developers building a CEX arbitrage bot with sub-100ms execution latency - cross-language coordination (Python orchestration, Go hot path) where every millisecond was the difference between profit and a missed window
- Deployed single-node k3s on NixOS with declarative config, Terraform provisioning, and GitOps workflow - reproducible infra, entire cluster from zero in under an hour
- Provisioned Kubernetes clusters on Yandex Cloud and Hetzner using Terraform for multiple clients - standardized the infra setup, cut provisioning time across engagements
- 100% completion rate across 8+ freelance engagements. Repeat clients. Handled everything end-to-end - from initial scoping call to production handoff

Tech: Python, Rust, JS/TS, Web3.py, ethers.js, PostgreSQL, Redis, asyncio, NixOS, k3s, Terraform, TimescaleDB, Docker

PROFESSIONAL DEVELOPMENT

Machine Learning Specialization - Stanford University (via Coursera) (2026)
Reinforcement Learning for Trading Strategies - New York Institute of Finance (2026)
Using Machine Learning in Trading and Finance - New York Institute of Finance (2026)

LANGUAGES

Russian: Native | English: Upper-Intermediate (B2)